EE2211 Introduction to Machine Learning

Lecture 1

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E7 05-01-14

Office Hour (Hybrid): Monday 11:00am - 12:00nn
(Week 2-4, Week 11-13)

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Course Contents

- Introduction and Preliminaries (Yueming)
  - ~~Introduction~~
  - Data Engineering
  - Introduction to Probability and Statistics
- Fundamental Machine Learning Algorithms I (Shaojing)
  - Systems of linear equations
  - Least squares, Linear regression
  - Ridge regression, Polynomial regression
- Fundamental Machine Learning Algorithms II (Shaojing)
  - Over-fitting, bias/variance trade-off
  - Optimization, Gradient descent
  - Decision Trees, Random Forest
- Performance and More Algorithms (Yueming)
  - Performance Issues
  - K-means Clustering
  - Neural Networks

For Yueming's part there will no coding questions in midterm and final

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World’s Largest Selfie

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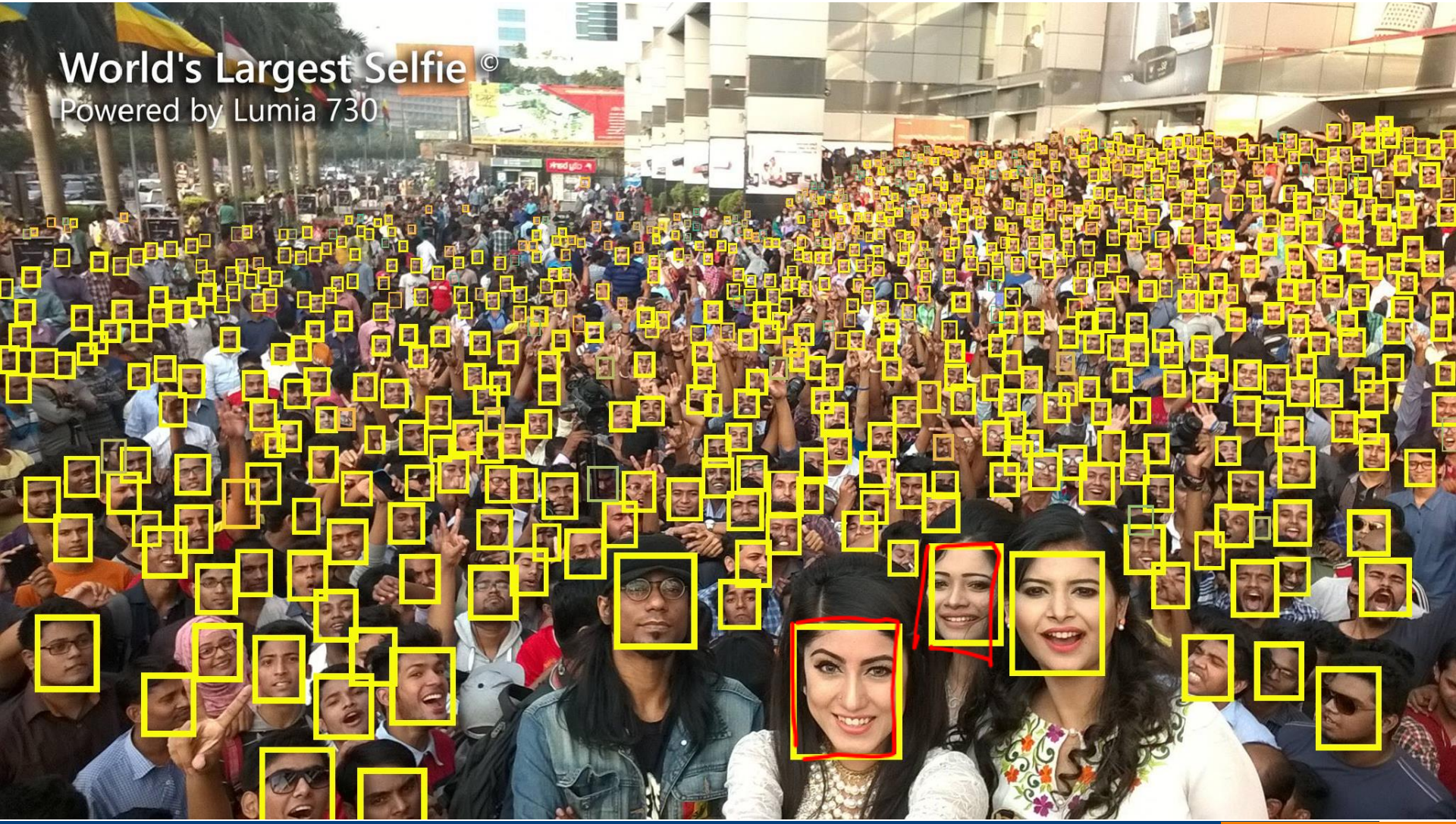
World’s Largest Selfie



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Recent Advances

ChatGPT 3.5 Sign up

You

write a poem about Singapore

LLM

ChatGPT

In a city-state where dreams ignite,
Singapore gleams with vibrant light,
Skyscrapers soar, a modern maze,
Where history meets tomorrow's craze.

Gardens bloom in meticulous grace,
Supertrees whisper, a futuristic embrace,
Marina Bay sparkles, a jewel aglow,
Reflecting a nation's spirit, in ebb and flow.

From hawker stalls to haute cuisine,
A melting pot where cultures convene,
Peranakan hues and Tamil beats,
Chinatown's bustle, each story repeats.

Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.

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Outline

- What is machine learning?
  - Three Definition(s)
- When do we need machine learning?
  - Sometimes we need, sometimes we don't
- Applications of machine learning
- Types of machine learning
  - Supervised, Unsupervised, Reinforcement Learning
- Walking through a toy example on classification
- Inductive vs. Deductive Reasoning

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What is machine learning?

Learning is any process by which a system improves performance from experience.

- Herbert Simon

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- Herbert Simon

A computer program is said to learn

- from ***experience E***
- with respect to some class of ***tasks T***
- and performance ***measure P***,

if its performance at tasks in T, as measured by P, improves with experience E.

- Tom Mitchell

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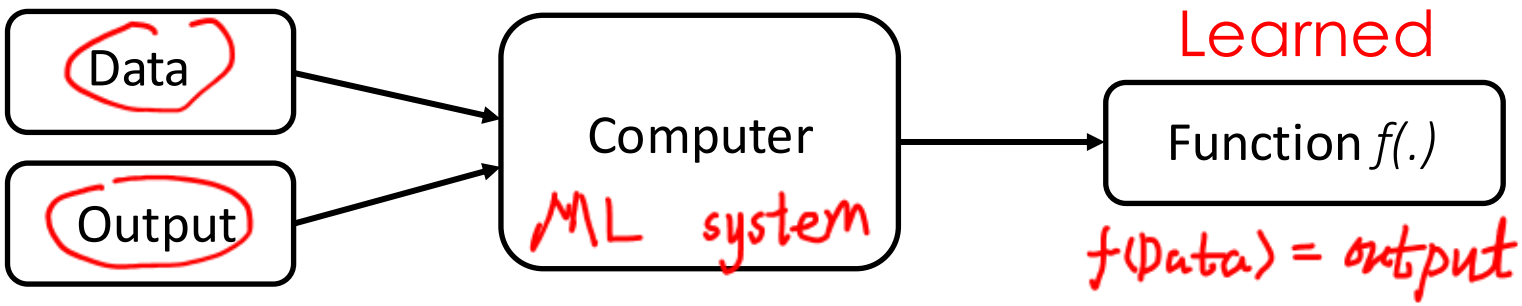
Machine Learning (Supervised Learning)



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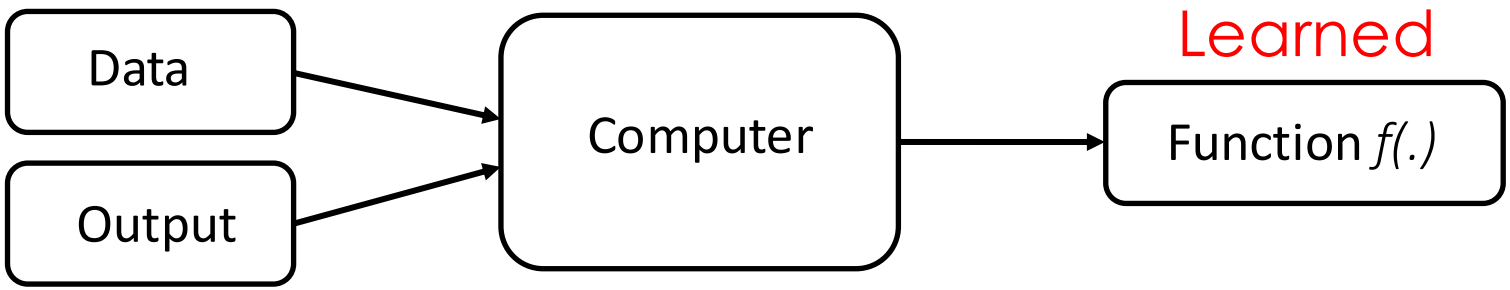
Machine Learning (Supervised Learning)



Data **Output** (label)

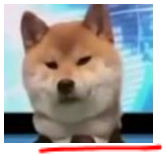
Cat

⋮ 10.000

Dog

$\boxed{f(.)}$ such that

$f($ $) =$ 'cat'

$f($  $) =$ 'dog'

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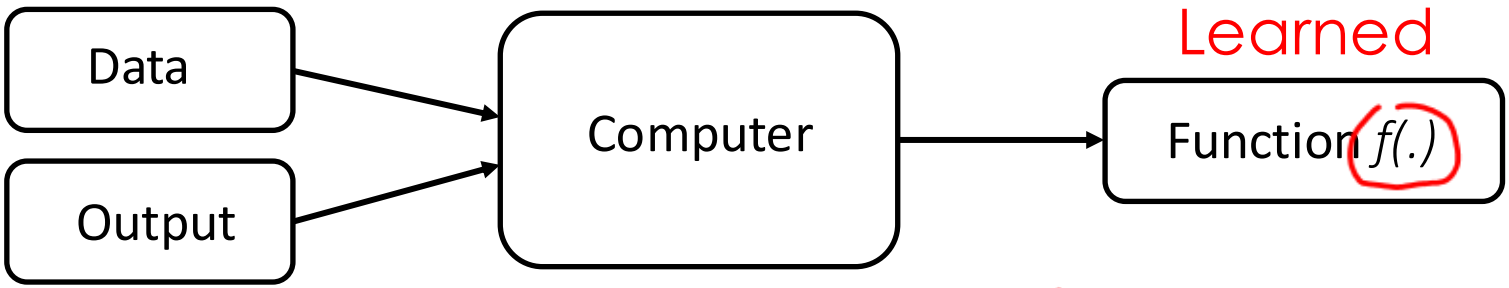
Machine Learning (Supervised Learning)



classification

Data **Output**

Cat

⋮

Dog

$f(.)$

training

When applied

$f($ $)$ → Cat !

New image

testing / inference

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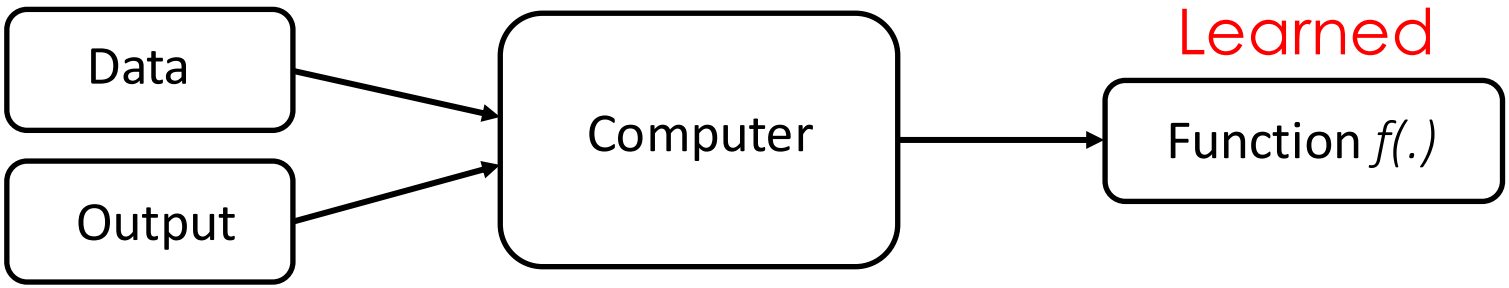
Machine Learning (Supervised Learning)



Data **Output**

Cat

⋮

Dog

$\longrightarrow f(\,.\,)$

When applied

$f($ $) \longrightarrow$ Cat !

New image

Machine Learning: field of study that gives computers the ability to learn without being explicitly programmed

- Arthur Samuel

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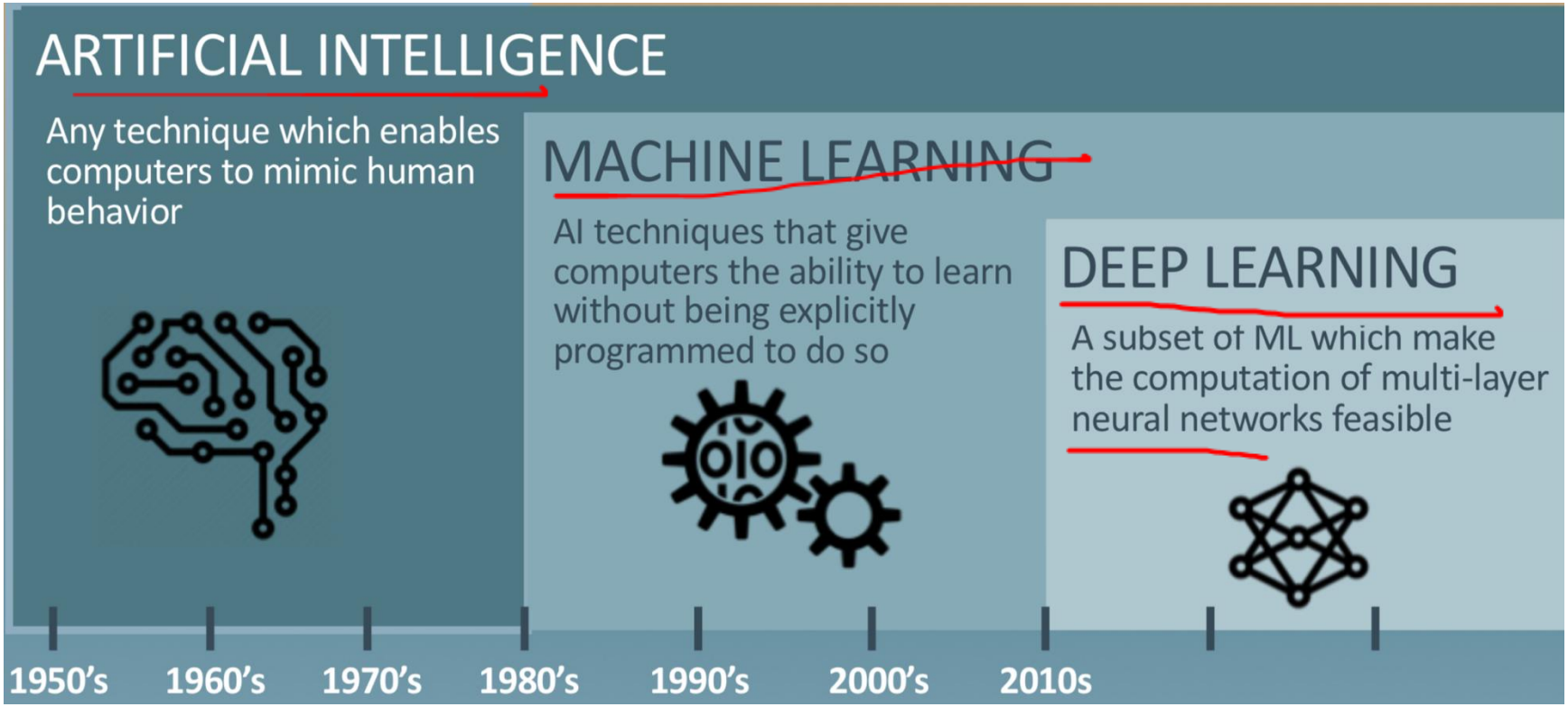
AI, Machine Learning, and Deep Learning



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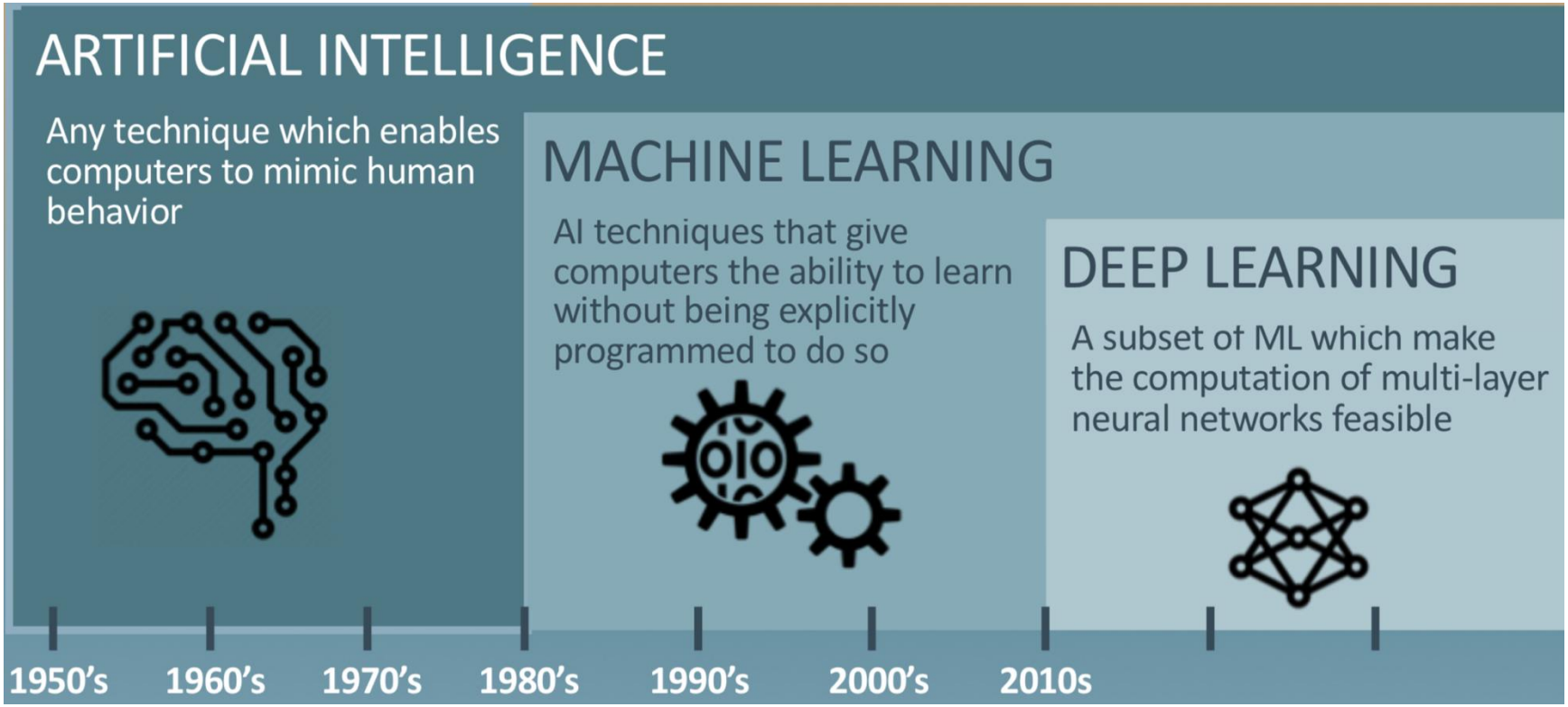
AI, Machine Learning, and Deep Learning



Example of AI but not ML: Deductive Reasoning

NUS is in Singapore, Singapore is in Asia -> NUS is in Asia

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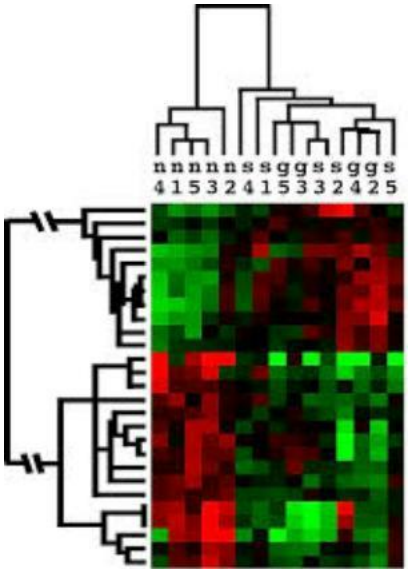
When do we need machine learning?

Lack of human expertise
(Navigating on Mars)

Involves huge amount of data
(Genomics)



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(Navigating on Mars)

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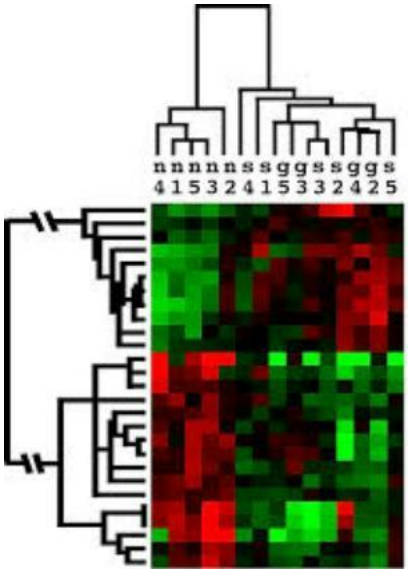

Learning is not always useful:

No need to "learn" to calculate payroll!

My Salary = Days_of_work * Daily Salary + Bonus

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Application of Machine Learning

Task T, Performance P, Experience E

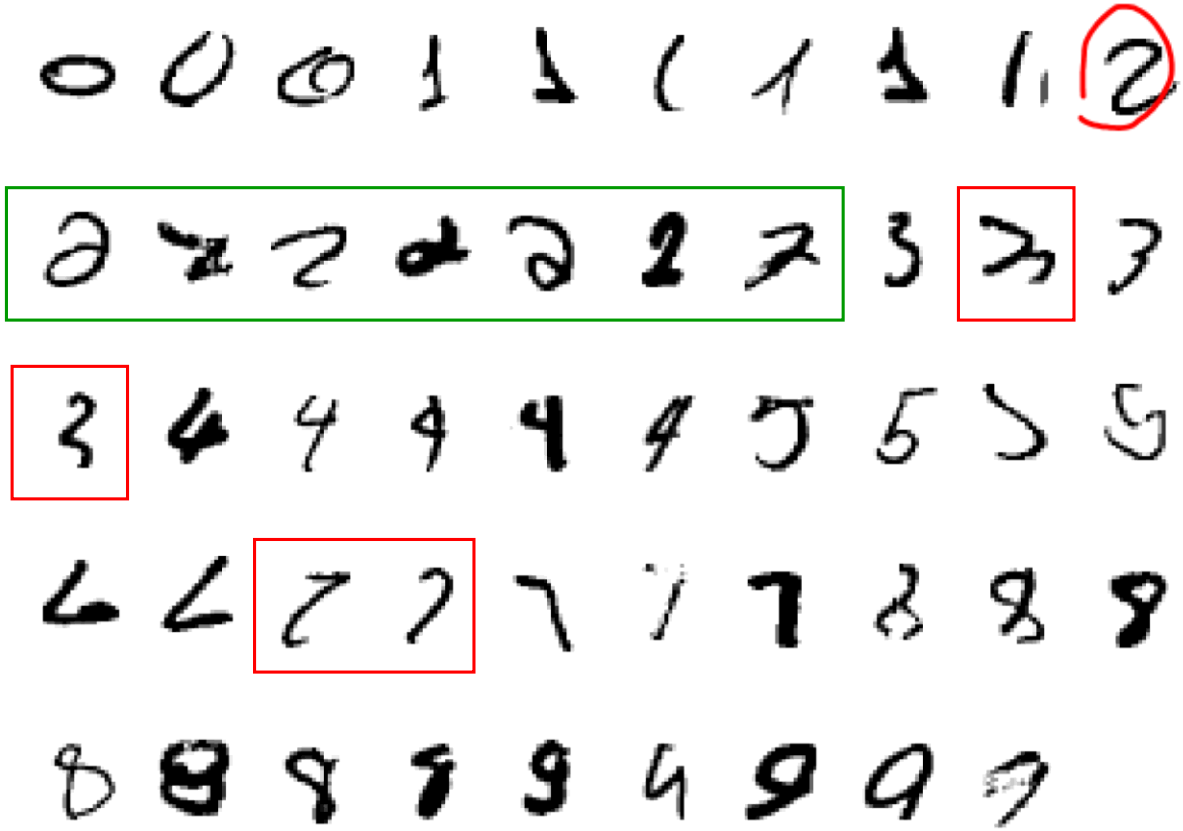
0 – 9

T: Digit Recognition
P: Classification Accuracy
E: Labelled Images

data output (label)

"four"

 "three"



Labels -> Supervision!

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Application of Machine Learning

Task T, Performance P, Experience E

T: Email Categorization
P: Classification Accuracy
E: Email Data, Some Labelled

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Application of Machine Learning

Task T, Performance P, Experience E

T: Playing Go Game
P: Chances of Winning
E: Records of Past Games

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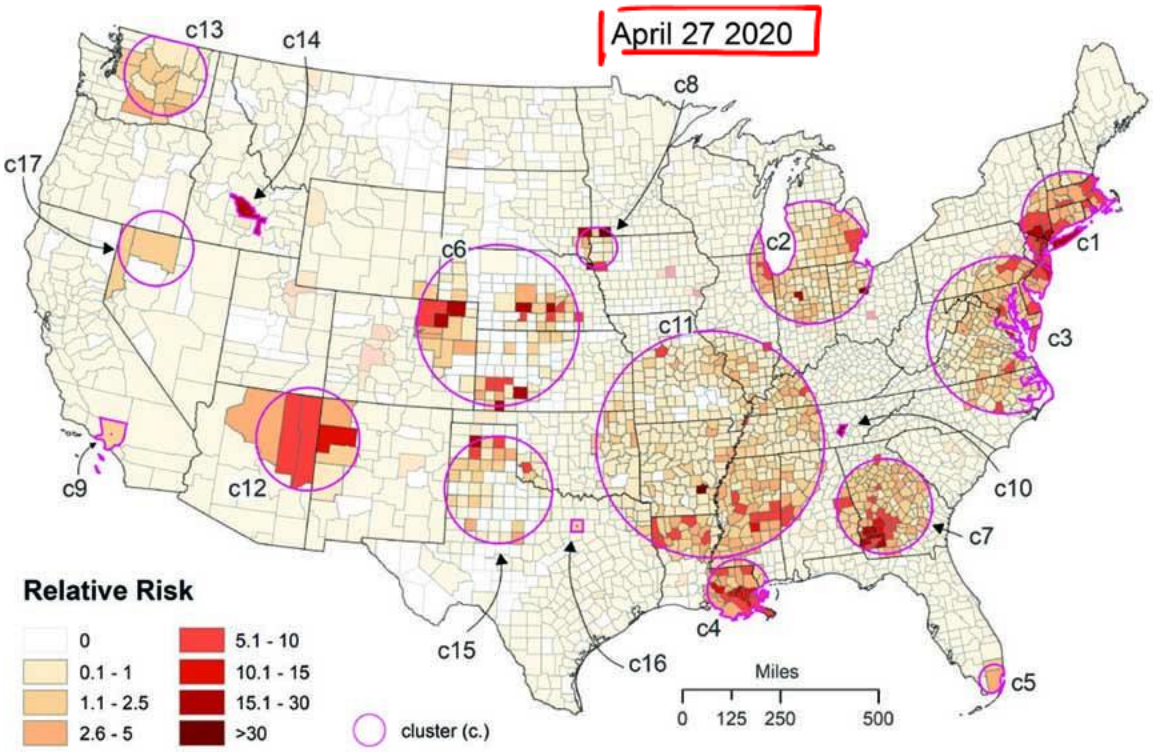
Application of Machine Learning

Task T, Performance P, Experience E

covid-19

T: Identifying Covid-19 Clusters
P: Small Internal Distances
Larger External Distances
E: Records of Patients



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Application of Machine Learning

Task T, Performance P, Experience E

T: Identifying Covid-19 Clusters
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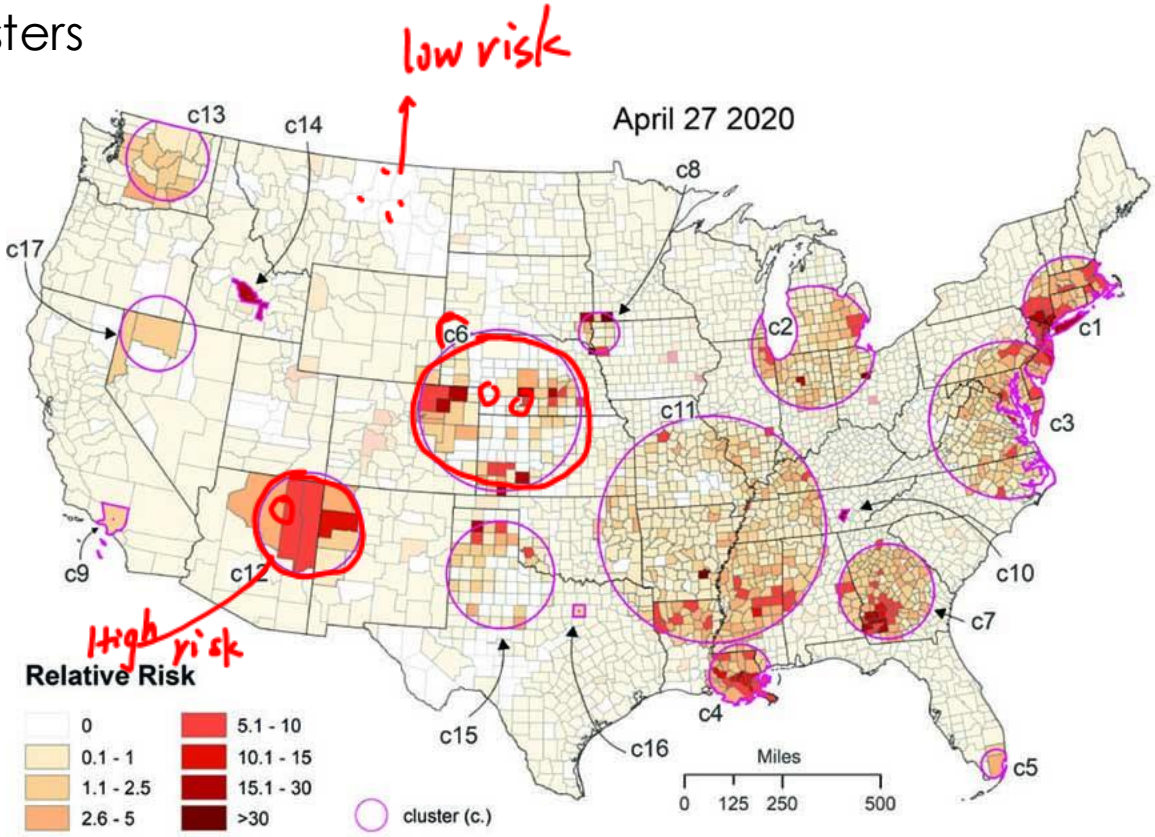

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Web Search Engine

Product Recommendation

Language Translation

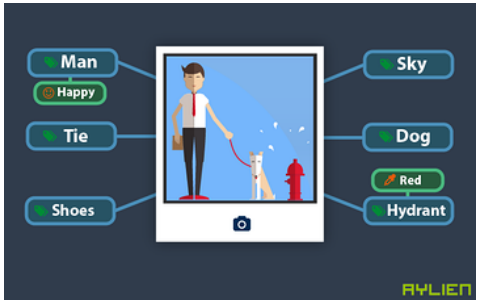
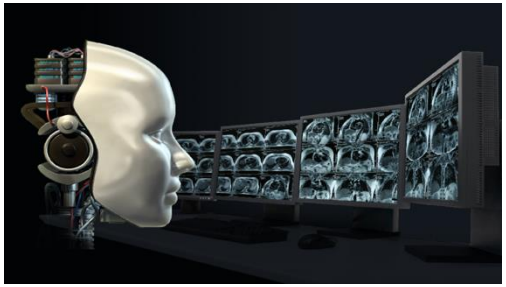


Photo Tagging

Virtual Personal Assistant

Portfolio Management

Traffic Prediction



Medical Diagnosis

Algorithmic Trading

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Types of Machine Learning

Supervised Learning

Input:
1) Training Samples,
2) Desired Output (Teacher/Supervision)

Output:
A rule that maps input to output

Unsupervised Learning

Input:
Samples

Output:
Underlying patterns in data

Reinforcement Learning

Input:
Sequence of States, Actions, and Delayed Rewards

Output:
Action Strategy: a rule that maps the environment to action

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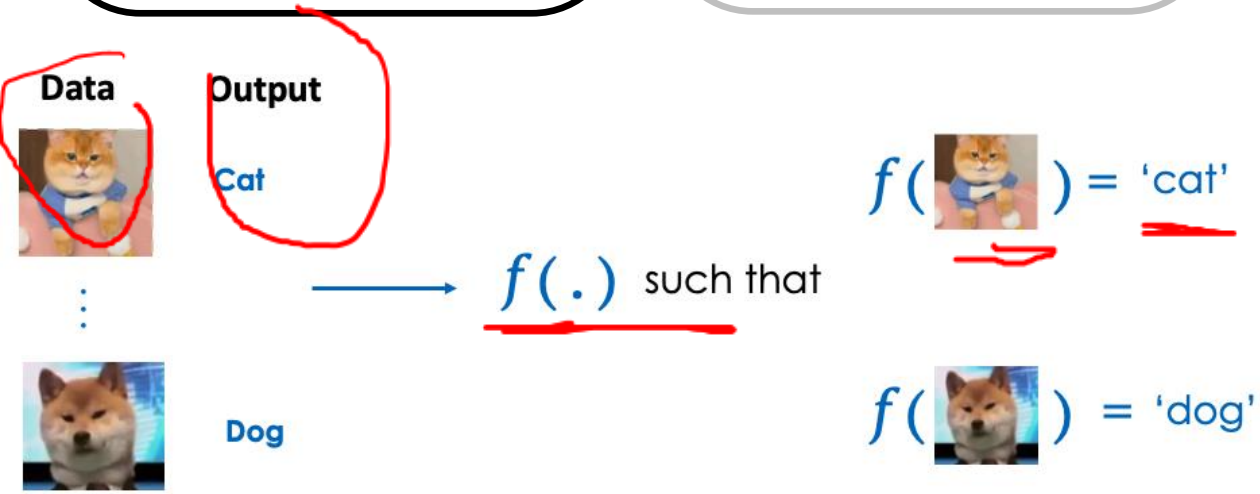

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Supervised Learning

x → Regression → y (Continuous)

x → Classification → y (Categorical)

label

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Supervised Learning

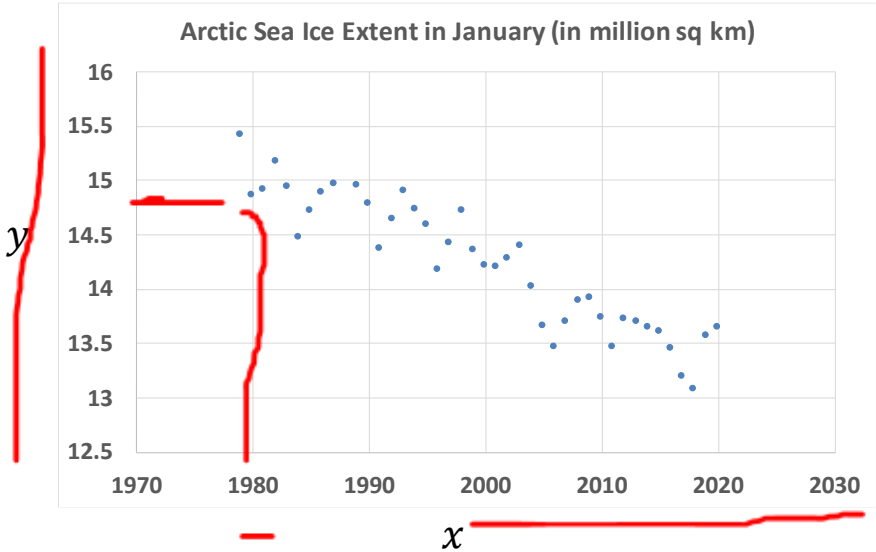
x → **Regression** → y (Continuous)

x → **Classification** → y (Categorical)

Regression

- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$
- Learn a function $f(\mathbf{x})$ to predict real-valued y given $\mathbf{x}$

data output



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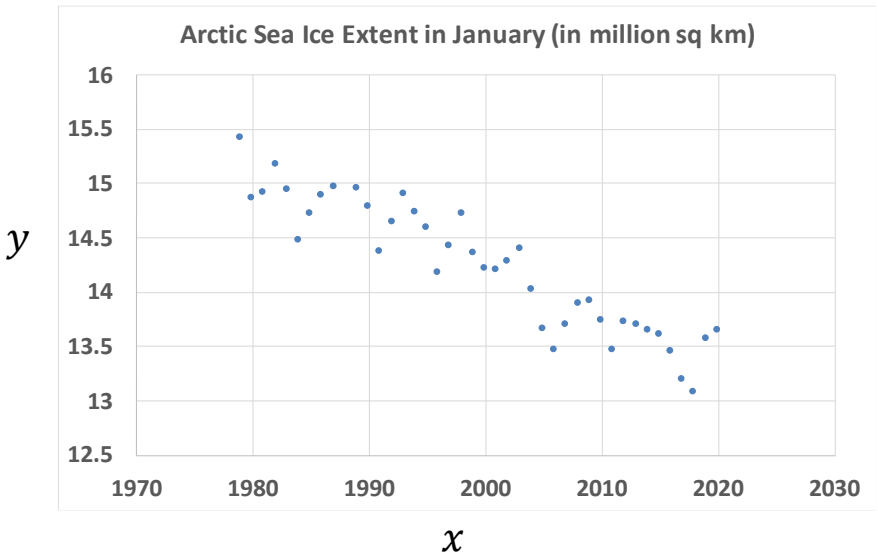

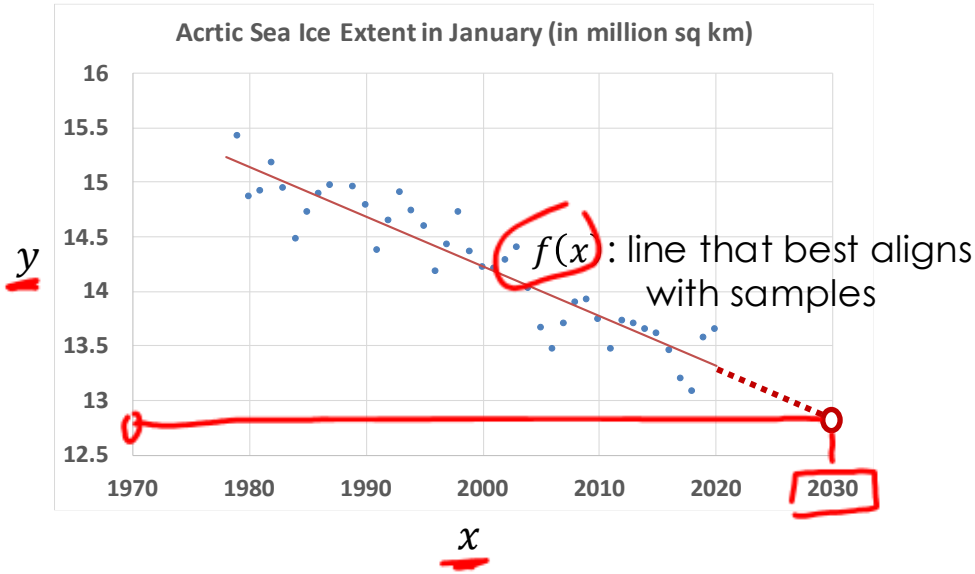

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Supervised Learning

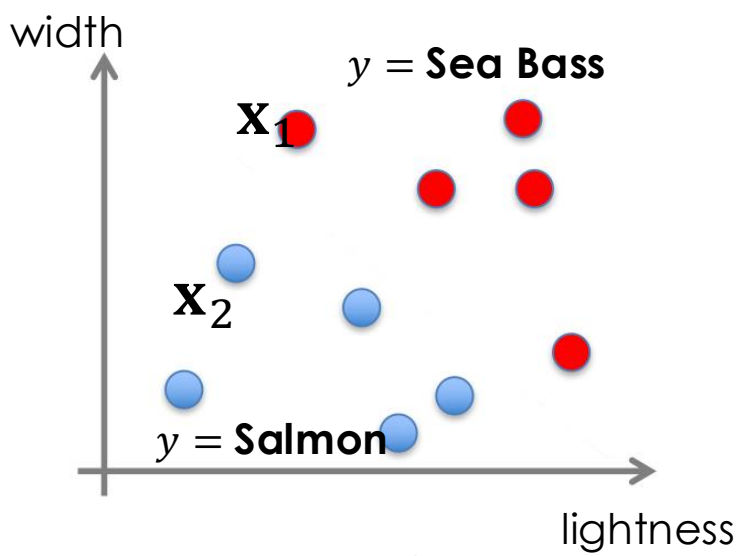
x → **Regression** → y (Continuous)

x → **Classification** → y (Categorical)

Classification

- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$
- Learn a function $f(\mathbf{x})$ to predict categorical y given $\mathbf{x}$

Feature Space



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Supervised Learning

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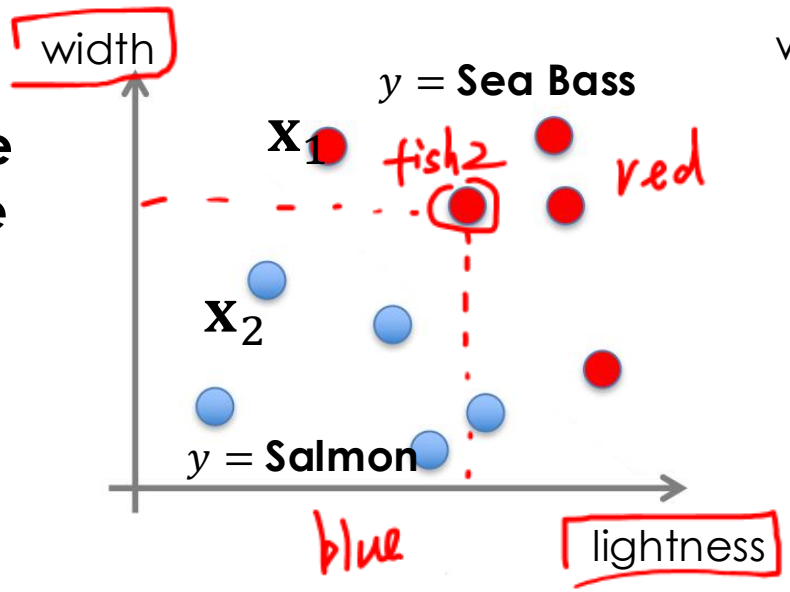

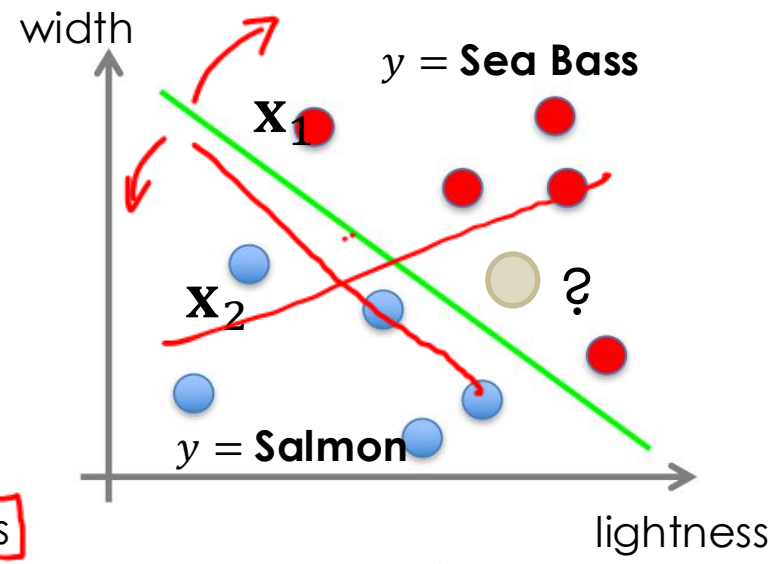

$f(x)$: line that separates two classes

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Types of Machine Learning

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Unsupervised Learning

Input:
Samples

Output:
Underlying patterns in data

Reinforcement Learning

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Sequence of States, Actions, and Delayed Rewards

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Action Strategy: a rule that maps the environment to action

Key different w.r.t. supervised learning
No Label/Supervision is given!

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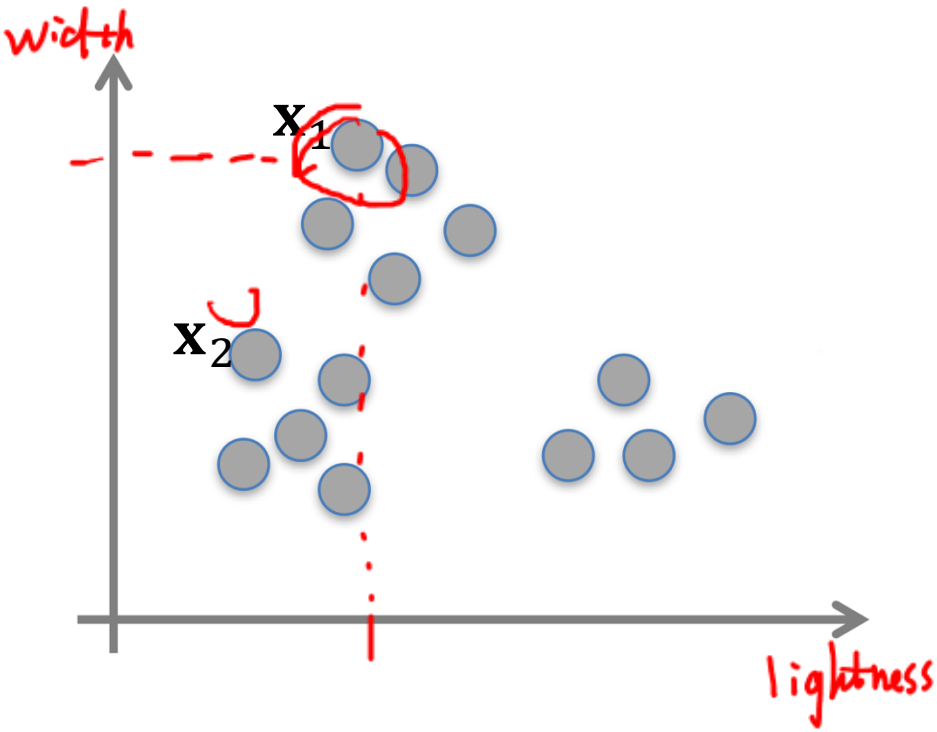
Unsupervised Learning

Clustering

- Given $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N$, without labels
- Output Hidden Structure Behind



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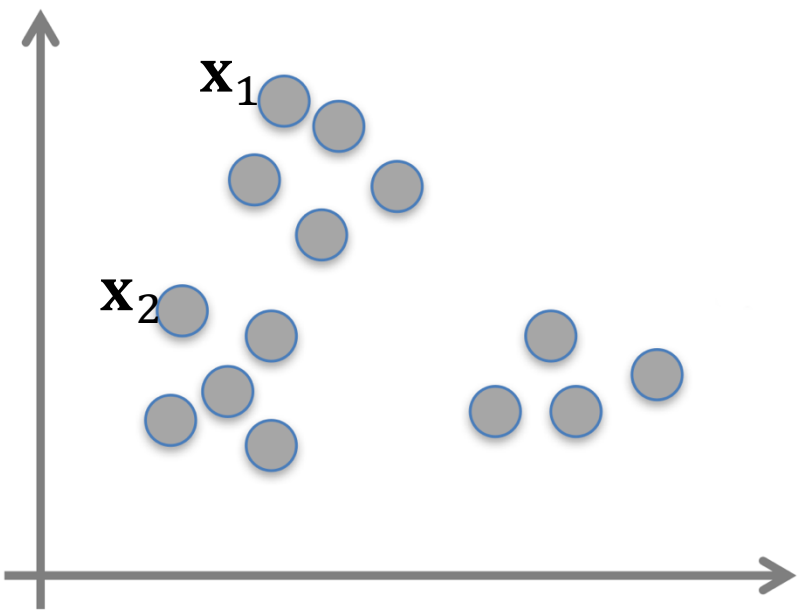
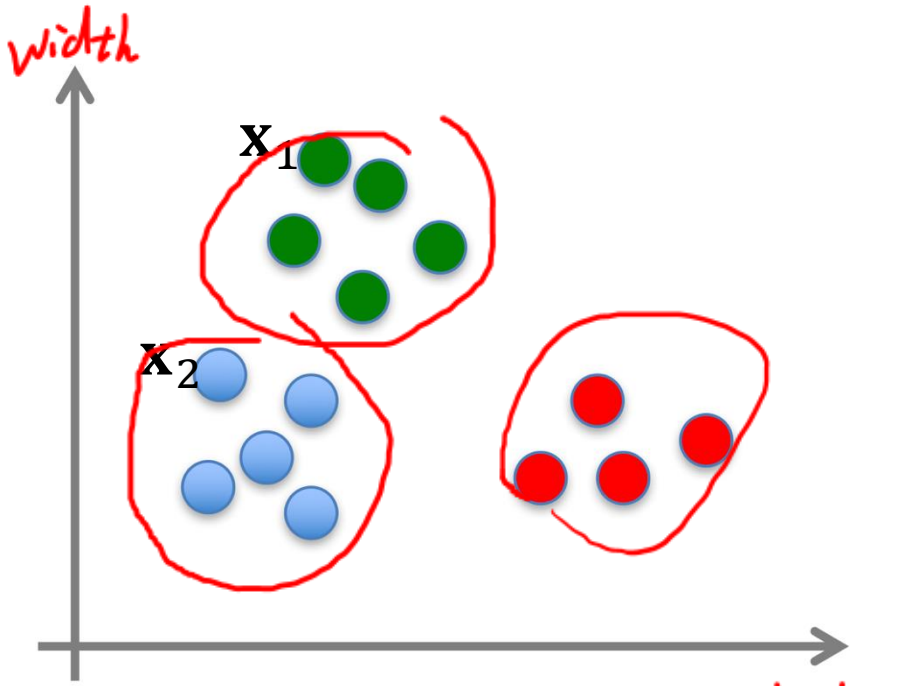
Unsupervised Learning

Grouping

Clustering

- Given $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N$, without labels
- Output Hidden Structure Behind





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Reinforcement Learning

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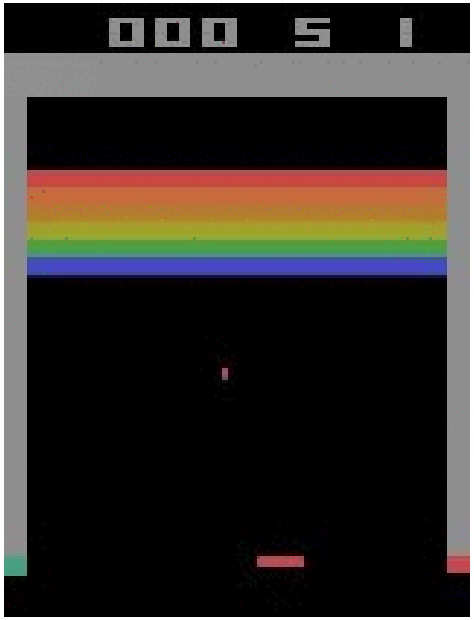
Reinforcement Learning

Breakout Game



Training 15 minutes

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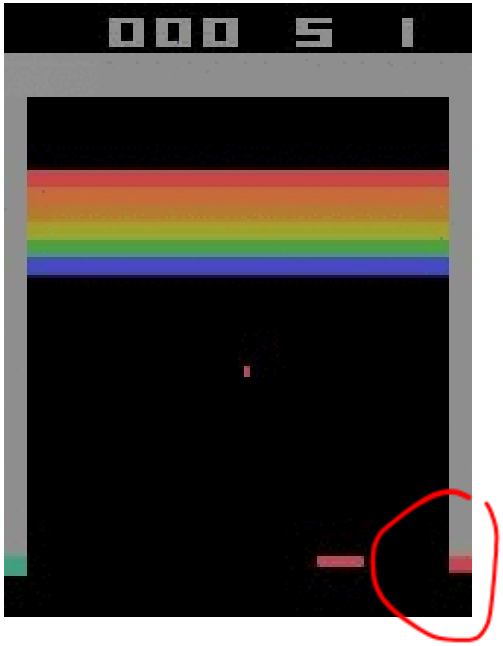
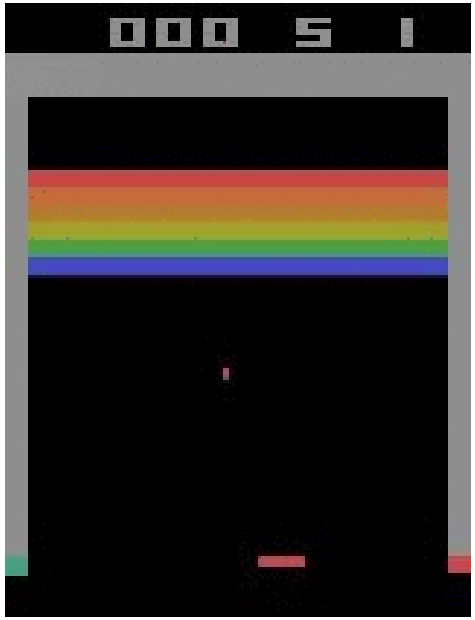
Reinforcement Learning

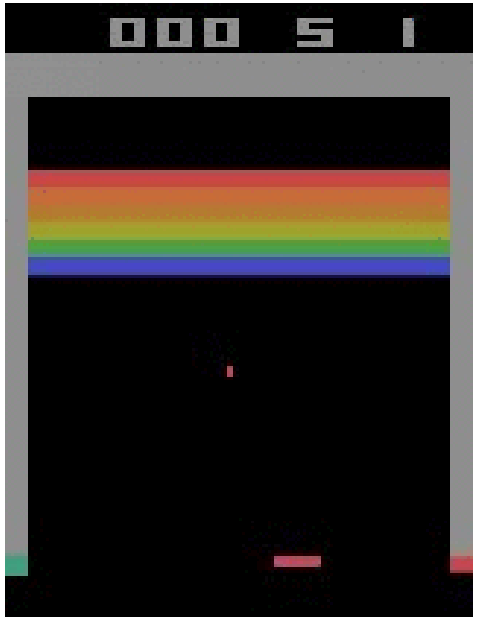
Breakout Game



Initial Performance



Training 15 minutes



Training 30 minutes

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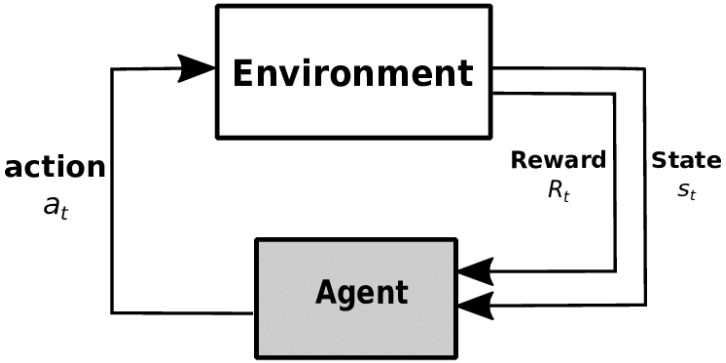
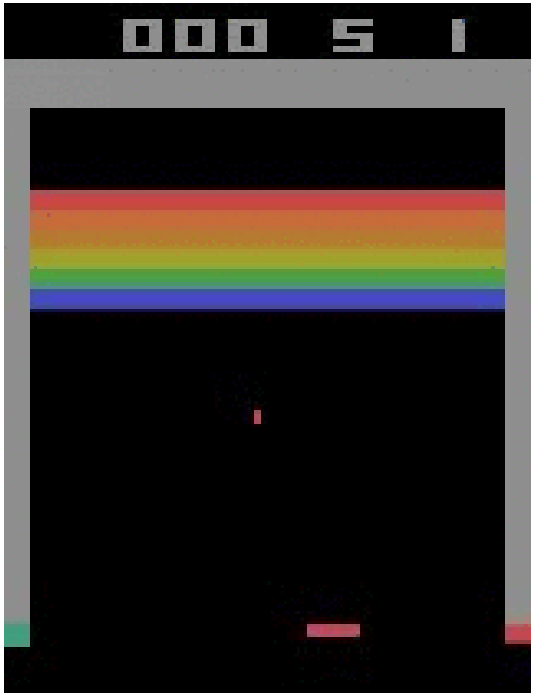
Reinforcement Learning

- Given sequence of states $\boldsymbol{S}$ and actions $\boldsymbol{A}$ with (delayed) rewards $\boldsymbol{R}$
- Output a policy $\pi(a, s)$, to guide us what action a to take in state s





$\boldsymbol{S}$: Ball Location, Paddle Location, Bricks

$\boldsymbol{A}$: left, right

$\boldsymbol{R}$:

positive reward
Knocking a brick, clearing all bricks

negative reward
Missing the ball

zero reward
Cases in between

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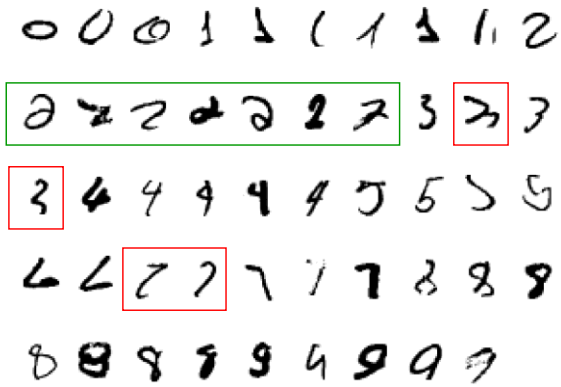
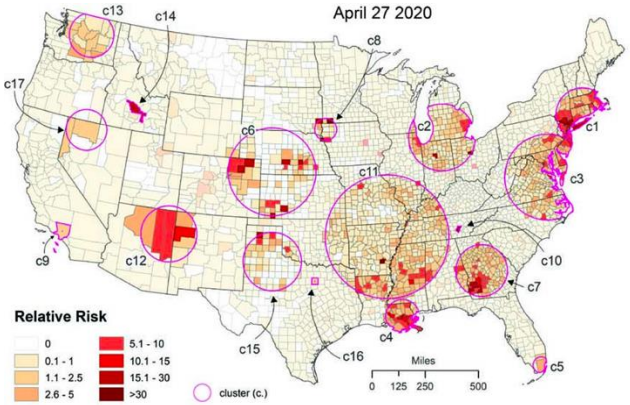
Supervised
Unsupervised
Reinforcement

Quiz Time!



Supervised



Unsupervised

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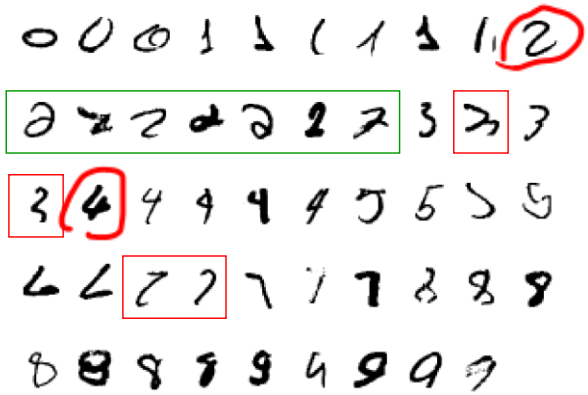
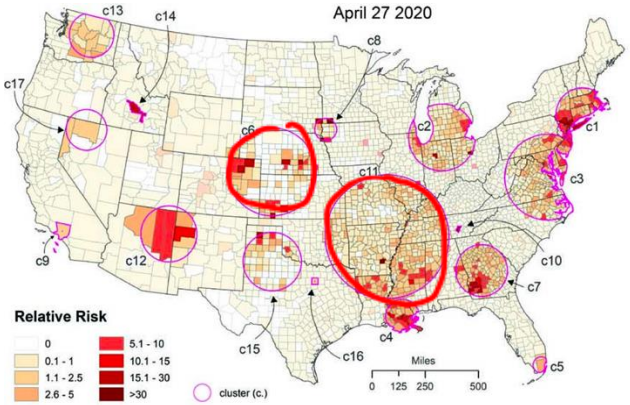
Supervised
Unsupervised
Reinforcement

Quiz Time!



Supervised



Unsupervised

Supervised

Reinforcement

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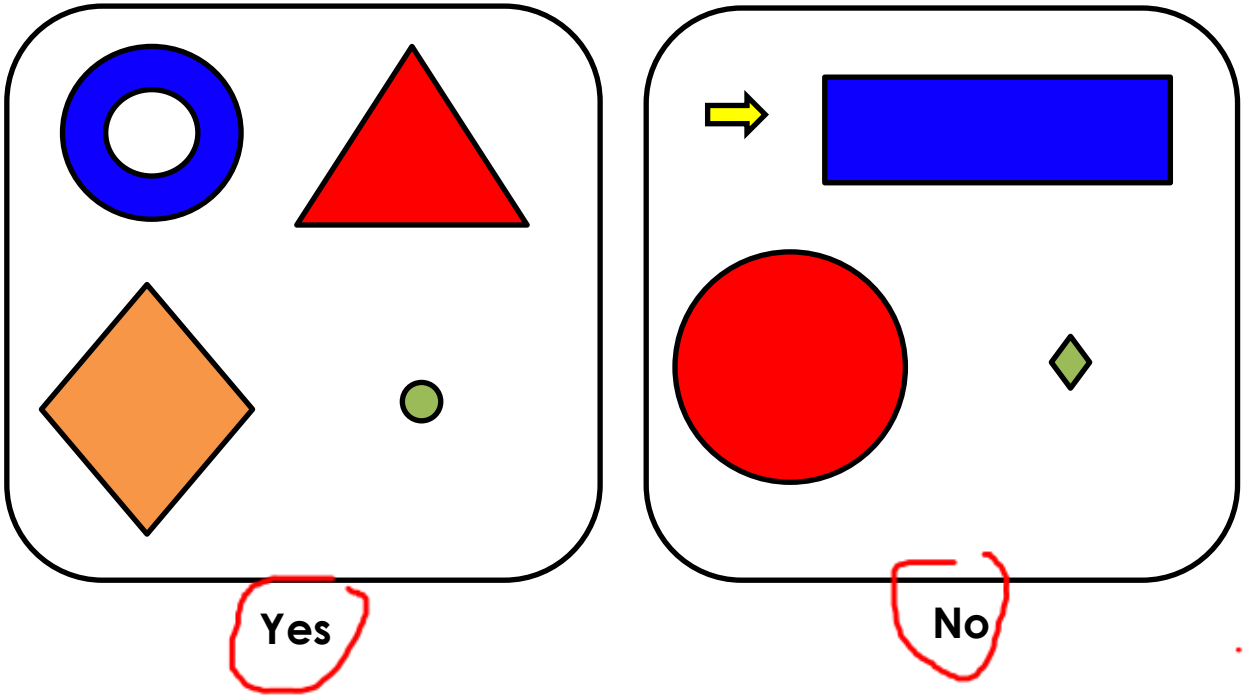
Walking Through A Toy Example: Token Classification



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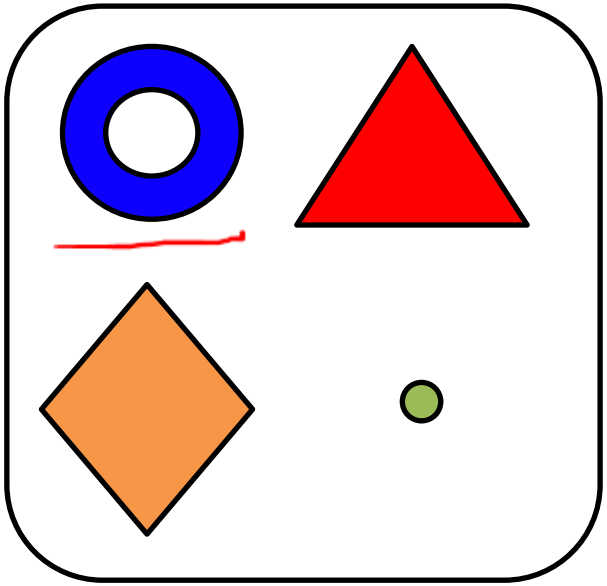
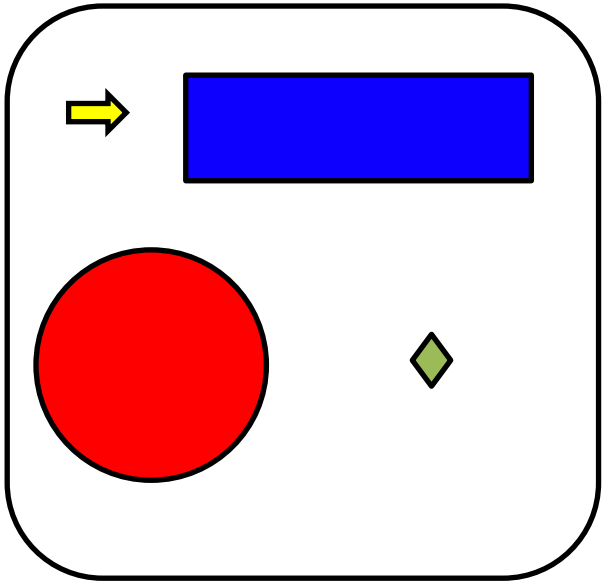
Walking Through A Toy Example: Token Classification

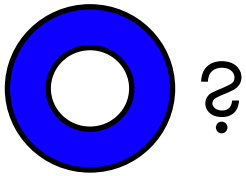


Yes

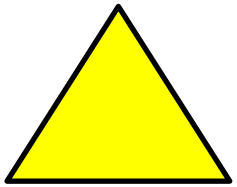


No

testing sample



? Yes



?

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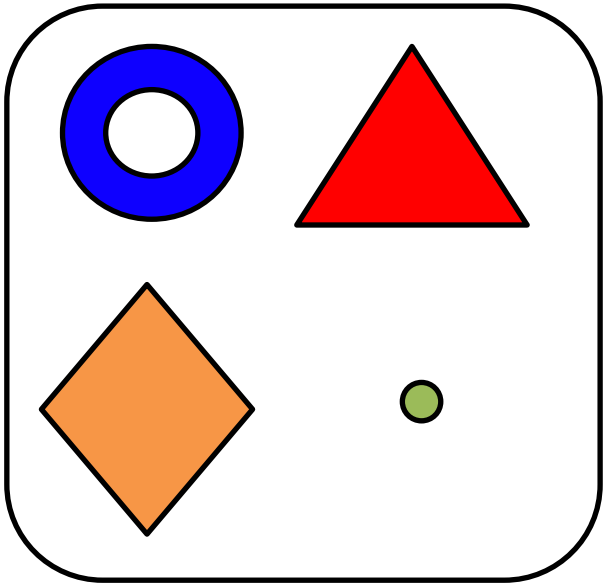
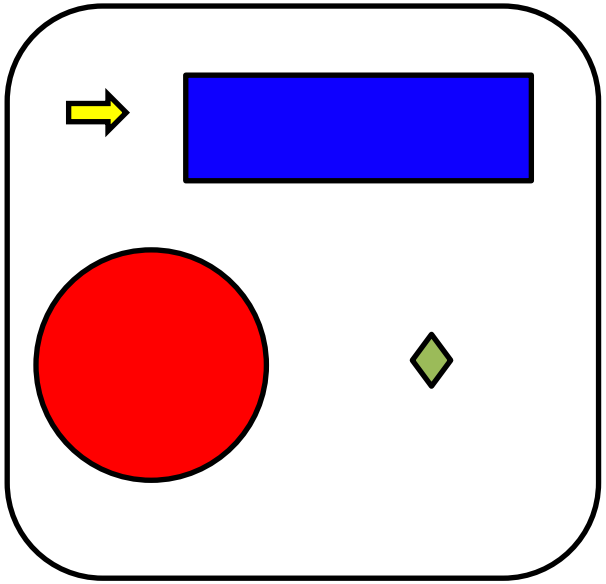
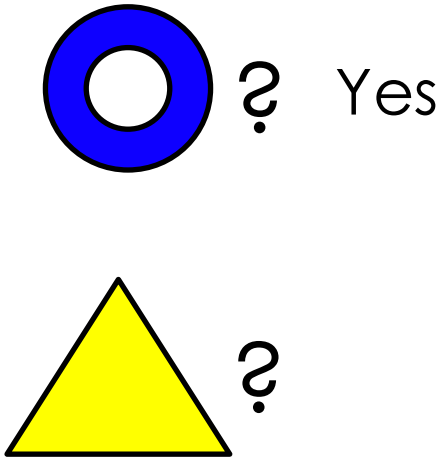
Walking Through A Toy Example: Token Classification



Yes



No



Step1: Feature Extraction
Extract Attributes of Samples

→

Step2: Sample Classification
Decide Label for a Sample

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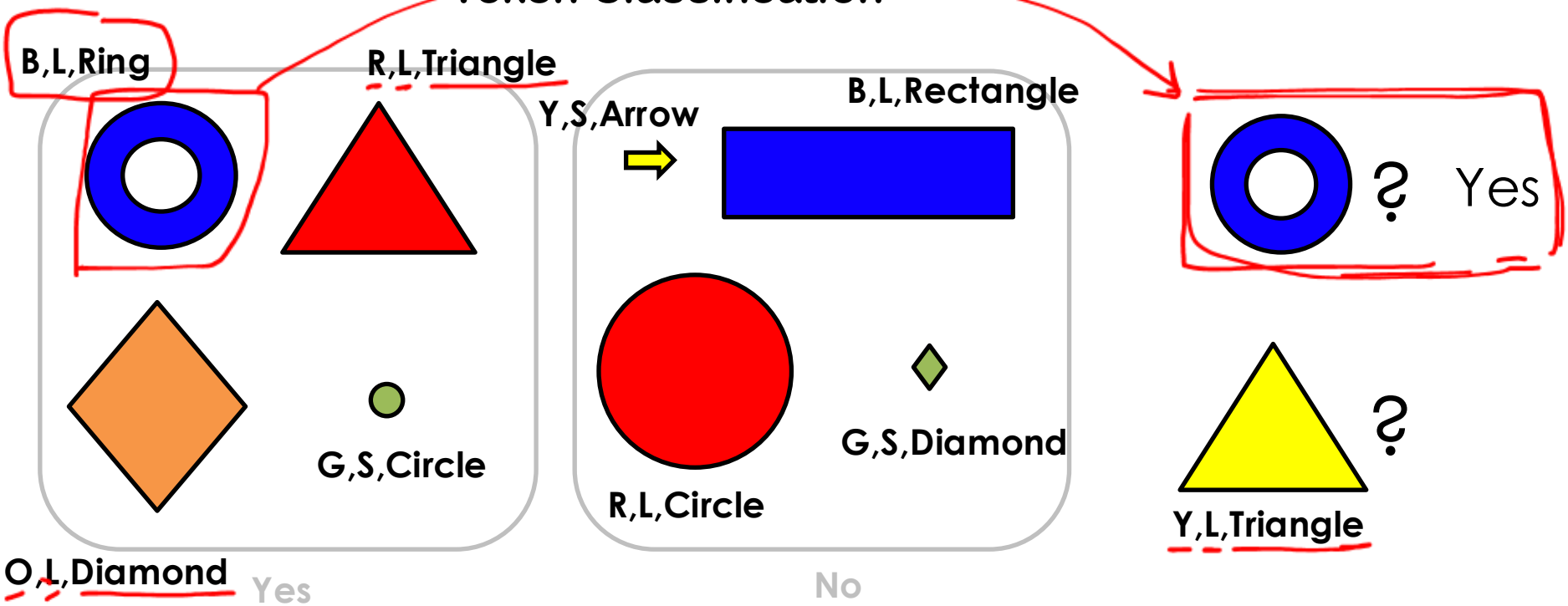
Walking Through A Toy Example: Token Classification



Step 1: Feature Extraction

Color, Size, Shape

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Walking Through A Toy Example: Token Classification

Feature Extraction

	Color	Size	Shape	Label
	Blue	Large	Ring	Yes
	Red	Large	Triangle	Yes
	Orange	Large	Diamond	Yes
	Green	Small	Circle	Yes
	Yellow	Small	Arrow	No
	Blue	Large	Rectangle	No
	Red	Large	Circle	No
	Green	Small	Diamond	No
	Yellow	Large	Triangle	?

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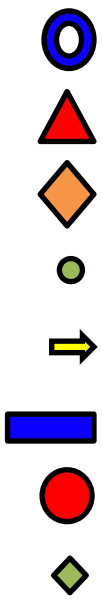
Walking Through A Toy Example: Token Classification

Feature Extraction



Color	Size	Shape	Label
Blue	Large	Ring	Yes
Red	Large	Triangle	Yes
Orange	Large	Diamond	Yes
Green	Small	Circle	Yes
Yellow	Small	Arrow	No
Blue	Large	Rectangle	No
Red	Large	Circle	No
Green	Small	Diamond	No

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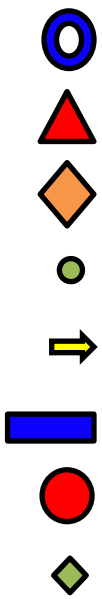
Walking Through A Toy Example: Token Classification

Feature Extraction



Color	Size	Shape	Label
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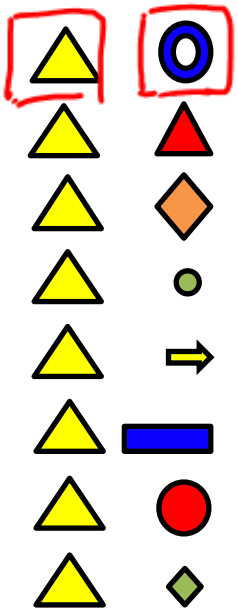
Red	Large	Triangle	Yes
Orange	Large	Diamond	Yes
Green	Small	Circle	Yes
Yellow	Small	Arrow	No
Blue	Large	Rectangle	No
Red	Large	Circle	No
Green	Small	Diamond	No

Similarity



Color	Size	Shape	Total
0	1	0	1
0	1	1	2
0	1	0	1
0	0	0	0
1	0	0	1
0	1	0	1
0	1	0	1
0	0	0	0

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Walking Through A Toy Example: Token Classification

Feature Extraction

	Color	Size	Shape	Label
	Blue	Large	Ring	Yes
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	Green	Small	Diamond	No

Similarity

		Color	Size	Shape	Total
		0	1	0	1
		0	1	1	2
		0	1	0	1
		0	0	0	0
		1	0	0	1
		0	1	0	1
		0	1	0	1
		0	0	0	0

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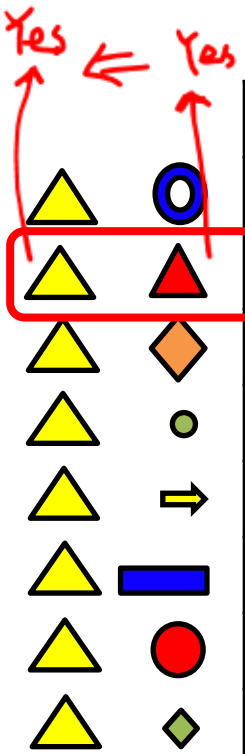
Walking Through A Toy Example: Token Classification



Similarity

Color	Size	Shape	Total
0	1	0	1
0	1	1	2
0	1	0	1
0	0	0	0
1	0	0	1
0	1	0	1
0	1	0	1
0	0	0	0

Nearest Neighbor Classifier:

1) Find the "nearest neighbor" of a sample in the feature space

2) Assign the label of the nearest neighbor to the sample

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Inductive vs. Deductive Reasoning

- Main Task of Machine Learning: to make inference

Two Types of Inference

Inductive

- To reach **probable** conclusions
- Not all needed information is available, causing uncertainty

Deductive

- To reach logical conclusions **deterministically**
- All information that can lead to the correct conclusion is available

Probability and Statistics

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Inductive vs. Deductive Reasoning

- Main Task of Machine Learning: to make inference

Two Types of Inference

Inductive

ML

- To reach **probable** conclusions
- Not all needed information is available, causing uncertainty

Deductive

- To reach logical conclusions **deterministically**
- All information that can lead to the correct conclusion is available

Probability and Statistics

Rule-based reasoning

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Inductive Reasoning

Note: humans use inductive reasoning all the time and not in a formal way like using probability/statistics.

Ref: Gardener, Martin (March 1979). "MATHEMATICAL GAMES: On the fabric of inductive logic, and some probability paradoxes" (PDF). *Scientific American*. **234**

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Summary by Quick Quiz

Three Components in ML Definition

Task T, Performance P, Experience E

Two Types of Supervised Learning

Classification, Regression

Three Types of in ML

Supervised Learning
Unsupervised Learning
Reinforcement Learning

One Type of Unsupervised Learning

Clustering

Inductive and Deductive

Inductive: Probable
Deductive: Rule-based

Example of a Classifier Model

Nearest Neighbor Classifier

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Practice Question
(Type of Question to Expect in Exams)

Which of the following statement is true?

A. Nearest Neighbor Classifier is an example of unsupervised learning
B. Nearest Neighbor Classifier is an example of deductive learning
C. Nearest Neighbor Classifier is an example of feature extraction
D. None of the above is correct.

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